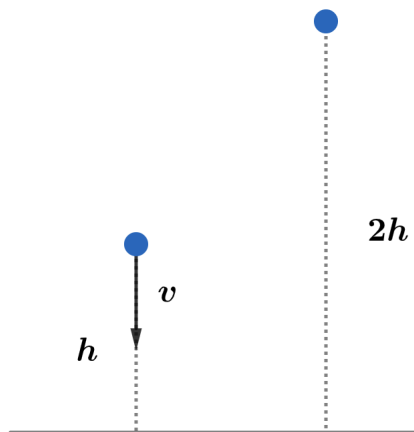


2020A F=ma Exam: Problem 1

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The ball has some initial energy E_i which is all in the form of kinetic energy right before it hits ground. Since it loses half of its kinetic energy after the bounce, we have from conservation of energy

$$\begin{aligned}\frac{E_i}{2} &= mg(2h) \\ E_i &= 4mgh\end{aligned}$$

Since $E_i = \frac{1}{2}mv_i^2 + mgh$,

$$\begin{aligned}\frac{1}{2}mv_i^2 + mgh &= 4mgh \\ \frac{1}{2}mv_i^2 &= 3mgh \\ v_i^2 &= 6gh \\ v_i &= \sqrt{6gh}\end{aligned}$$

so the answer is $\boxed{\text{E}}$.